

Name _____ Surname _____
Date ____/____/____

Marking Group X's thermometer report

The brief. A group of students (Group X) built a thermometer from a thermistor connected to Arduino. The client's requirement was: **accuracy of $\pm 1^\circ\text{C}$ between 0°C and 45°C** . They calibrated it in four water baths and checked it in a fifth, independent bath.

Your job today. You are the examiner. Group X's report has mistakes in it — some real, some things that only look wrong. Find them, using the criteria below, then finish the analysis they did not finish properly.

Part 1 — Marking criteria

A good calibration graph and a good results table should have:

- (a) axes labelled with the *quantity* and its *unit* (e.g. $T / ^\circ\text{C}$);
- (b) a scale that uses more than half of the grid in both directions;
- (c) points plotted accurately;
- (d) a single, thin, ruled **line of best fit** — never a dot-to-dot line, never forced through every point;
- (e) values recorded to a precision that matches the resolution of the instrument used;
- (f) repeated readings, so an anomalous result can be identified and excluded.

Part 2 — Group X's data table

Group X used a thermistor voltage divider (reading N , a number from 0 to 1023) and a reference liquid-in-glass thermometer with 1°C divisions. They took five thermistor readings in each of four calibration baths.

Bath	Reference T ($^\circ\text{C}$)	Thermistor reading, N				
1 – ice water	0.5	206	208	205	207	206
2 – cold water	12.0	296	298	254	297	299
3 – room temperature	22.03	378	380	377	379	381
4 – hot water	43.25	552	555	550	553	551

Note on bath 4: the reference thermometer read 45.0°C at the start of the readings and 41.5°C at the end. The value 43.25°C is their average.

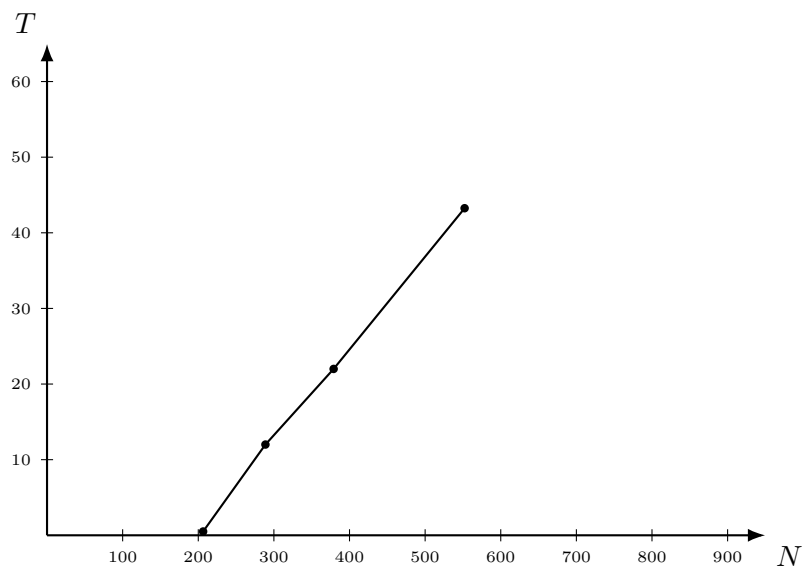
Task. Using the criteria in Part 1, find **at least four** things in this table that a real problem, and **at least one** thing that looks like a problem but is not. For each one, say which criterion is involved.

Real problems:

Not actually a problem (explain why):

Part 3 — Group X's graph

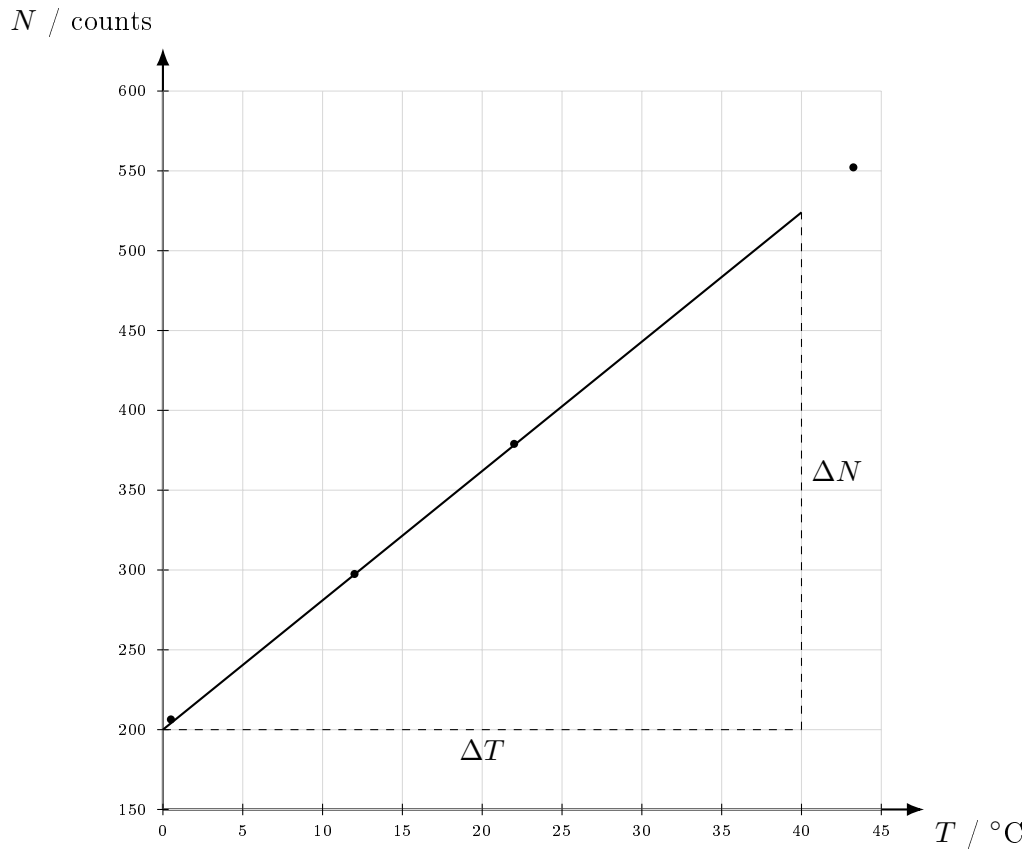
This is the graph Group X drew from their table.



Task. List at least four things wrong with this graph, using the criteria from Part 1.

Part 4 — The corrected graph

Here is the same data, plotted correctly.



Task.

- (a) Read the gradient triangle. Calculate the gradient m (in counts per $^{\circ}\text{C}$).

- (b) Read the intercept N_0 (the value of N where the line crosses $T = 0$).

- (c) Write the calibration equation in the form $T = \frac{N - N_0}{m}$.

Part 5 — The test bath

Group X checked their thermometer in a fifth bath, independent of the four calibration baths.

Reading in the test bath $N = 444$

Task.

- (a) What is wrong with how this single measurement was taken, compared with the four calibration baths?

- (b) Using your equation from Part 4, calculate the temperature of the test bath, T_{test} .

- (c) Your teacher will now give you the temperature of the test bath measured with the reference thermometer, T_{ref} . Write it here and calculate the difference.

$T_{\text{ref}} =$ _____ $^{\circ}\text{C}$ $|T_{\text{ref}} - T_{\text{test}}| =$ _____ $^{\circ}\text{C}$

- (d) Does Group X's thermometer meet the brief ($\pm 1^{\circ}\text{C}$, between 0°C and 45°C)?

- (e) **Quick check.** Using only bath 1 and bath 4 (the two end baths), write a quick two-point equation and use it to calculate T_{test} again.

- (f) Compare the two answers you got for T_{test} . Which one do you trust more, and why?

Part 6 — Suggested improvements

For **four** of the problems you found in Group X's report, suggest one specific, concrete improvement each. A good improvement names an instrument or a procedure – “be more careful” is not a valid answer.

1. _____

2. _____

3. _____

4. _____